

Counting the regions of the Clebsch cubic minus its 27 lines

1 The question

Let $X = V(f_3) \subset \mathbb{P}^3$ be the Clebsch cubic surface and let g be the nonic with $V(g) \cap X$ equal to the 27 lines. The routing function $r = g/q^5$, with $q > 0$ a shifted sum of squares, has

$$|r| \longrightarrow 0 \quad \text{on the 27 lines and as } \|x\| \rightarrow \infty,$$

so `connected_components` returns the connected components of $X_{\text{aff}} \setminus V(g)$, where $X_{\text{aff}} = X(\mathbb{R}) \cap \mathbb{R}^3$ is the affine part of the surface in the chosen chart.

Writing H_∞ for the plane at infinity and $C_\infty = X \cap H_\infty$, we have $X_{\text{aff}} = X(\mathbb{R}) \setminus C_\infty$, so

$$X_{\text{aff}} \setminus V(g) = X(\mathbb{R}) \setminus (27 \text{ lines} \cup C_\infty).$$

The count we want is therefore the number of *faces* of the curve arrangement $\mathcal{A} = \{27 \text{ lines}\} \cup C_\infty$ on the closed surface $X(\mathbb{R})$. The arrangement is a CW decomposition, so Euler's formula gives

$$F = \chi(X(\mathbb{R})) - V + E. \tag{1}$$

Everything below is the computation of V and E .

2 The Euler characteristic of $X(\mathbb{R})$

All 27 lines of the Clebsch surface are real. A smooth real cubic surface whose 27 lines are all real is $\mathbb{RP}^2$ blown up at six real points. A real blow-up replaces a point by an $\mathbb{RP}^1$, and $\chi(\mathbb{RP}^1) = 0$, so each blow-up changes the Euler characteristic by -1 . Hence

$$\chi(X(\mathbb{R})) = \chi(\mathbb{RP}^2) - 6 = 1 - 6 = -5. \tag{2}$$

3 The arrangement of the 27 lines

Every line on a smooth cubic surface meets exactly 10 of the others, so the number of intersecting pairs is

$$\frac{1}{2} \cdot 27 \cdot 10 = 135.$$

A general cubic surface

If no three lines are concurrent, those 135 pairs give 135 distinct points and each line carries 10 of them. A line of $X(\mathbb{R})$ is a circle, and k marked points cut a circle into k arcs, so

$$V = 135, \quad E = 27 \cdot 10 = 270, \quad F = -5 - 135 + 270 = \boxed{130},$$

which is the number quoted for a general real cubic surface with all lines real.

The Clebsch surface

The Clebsch is not general: it carries 10 *Eckardt points*, where three of the 27 lines are concurrent. (It is the unique cubic surface with ten of them.) Each Eckardt point accounts for $\binom{3}{2} = 3$ of the 135 intersecting pairs but contributes only one point, so

$$V = 135 - 3 \cdot 10 + 10 = 115. \quad (3)$$

Let $e(\ell)$ be the number of Eckardt points on a line ℓ . Of the ten lines meeting ℓ , exactly $2e(\ell)$ of them meet it at Eckardt points, two at each such point; the remaining $10 - 2e(\ell)$ meet it at distinct simple points. So ℓ carries

$$e(\ell) + (10 - 2e(\ell)) = 10 - e(\ell)$$

marked points, hence that many arcs. Since each Eckardt point lies on three lines, $\sum_{\ell} e(\ell) = 3 \cdot 10 = 30$, and

$$E = \sum_{\ell} (10 - e(\ell)) = 270 - 30 = 240. \quad (4)$$

Substituting (2), (3) and (4) into (1),

$$F = -5 - 115 + 240 = \boxed{120}$$

faces in $\mathbb{RP}^3$. The ten Eckardt points cost exactly ten regions relative to the general surface.

4 The curve at infinity

The computation is affine, so C_{∞} must be added to the arrangement. It is a genuine extra curve: restricting the degree-three part of the defining equation to random real lines yields 1 or 3 real roots depending on the line, whereas three real lines would give three every time. So H_{∞} is *not* a tritangent plane and C_{∞} is not a union of three of the 27 lines.

Each line ℓ meets H_{∞} in a single point, which lies on X and hence on C_{∞} : twenty-seven incidences, but numerically only

$$21 \text{ distinct points,} \quad \text{of which 3 are already vertices.}$$

Those three are Eckardt points lying at infinity; they carry $3 \cdot 3 = 9$ of the lines between them. Hence:

Vertices. The new vertices are the points of $\ell \cap H_{\infty}$ not already counted in (3):

$$V' = 115 + (21 - 3) = 133. \quad (5)$$

Edges. A line gains one further marked point unless its point at infinity was already one of its marked points; the 9 lines through the three Eckardt points at infinity gain nothing, the other 18 gain one. The curve C_{∞} itself now carries the 21 marked points and so contributes 21 arcs.

$$E' = \underbrace{240}_{(4)} + \underbrace{18}_{\text{new marks}} + \underbrace{21}_{C_{\infty}} = 279. \quad (6)$$

(If some component of C_{∞} carried no marked point, it would need one dummy vertex *and* one dummy edge, leaving $V' - E'$ unchanged; the number of components of C_{∞} therefore does not affect the answer.)

Faces. By (1), (5) and (6),

$$F' = -5 - 133 + 279 = \boxed{141}.$$

5 Why 141 is also the number of routing points

Fix a face C of the arrangement. On C the function $|r|$ is positive, vanishes on the whole of ∂C — the boundary consists of arcs of the 27 lines, where $g = 0$, and of C_∞ , where $|r| \rightarrow 0$ because $\deg q^5 = 10 > 9 = \deg g$ — and is proper. It therefore attains an interior maximum, which is a critical point of $r|_X$, i.e. a routing point.

Morse theory gives $\chi(C) = \sum_p (-1)^{\text{index}(p)}$, summed over the critical points in C . Every face here is an open disc, so $\chi(C) = 1$ and generically C contains exactly one critical point, of index 0. Summing over faces predicts

141 routing points, every one of index 0.

Because all indices vanish, the connectivity matrix of `find_connectivity_matrix` is the identity: no gradient path is tracked, and the number of components is simply the number of critical points found.

6 Numerical confirmation

The combinatorial input was computed from the surface itself, and the routing points independently by the package.

Quantity	Predicted	Computed
real lines	27	27
intersecting pairs	135	135
distinct intersection points V	115	115
Eckardt points	10	10
lines with 8 / 10 marked points	15 / 12	15 / 12
arcs E	240	240
distinct points of $\ell \cap H_\infty$	—	21
of those, already vertices	—	3
lines gaining a marked point	—	18
faces F'	141	—
routing points found	—	141 , all of index 0

The two counts are obtained independently, so their agreement is mutual confirmation. Landing on exactly 141 rather than more also confirms, after the fact, the assumption that every face is a disc: an annular face would have contributed an extra critical point of index 1, and none was found.

7 Caveats

- The incidence counts (135 pairs, 10 Eckardt points, and the 21/3/18 figures at infinity) are numerical, decided with a tolerance of 10^{-6} on normalised homogeneous coordinates.

- C_∞ is assumed smooth, i.e. H_∞ is nowhere tangent to X . A tangency would change both V' and E' .
- The step from 141 faces to 141 critical points assumes each face is a disc carrying a single non-degenerate maximum of $|r|$; this is confirmed a posteriori by the computation.
- 141 counts faces in this particular affine chart. A different chart moves C_∞ and changes the answer; the chart-independent number is the projective one, 120.